

Midterm Solutions

ECON 403/503

Fall 2022



Problem 1

(a) The following knowledge space describes the situation:

$$N = \{A, B, U\}$$

$$\Omega = \{EEE, EES, ESE, SEE, ESS, SES, SSE, SSS\}$$

The player's information partitions & Type Spaces / signalling functions

$$P_A = \{\{EEE\}, \{EES\}, \{ESE\}, \{ESS\}, \{SEE, SSE, SES, SSS\}\}$$

$$T_A: \begin{matrix} \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ a & b & c & d & e \end{matrix} \quad \longleftrightarrow \Sigma_A$$

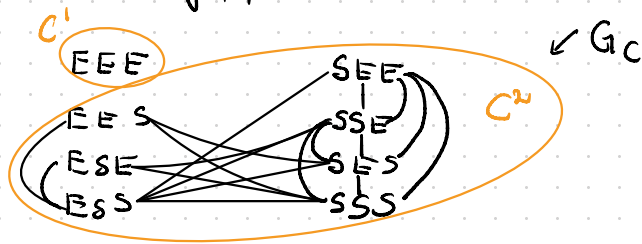
$$P_B = \{\{EEE\}, \{EES\}, \{SEE\}, \{SES\}, \{ESS, ESE, SSE, SSS\}\}$$

$$T_B: \begin{matrix} \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ a' & b' & c' & d' & e' \end{matrix} \quad \longleftrightarrow \Sigma_B$$

$$P_U = \{\{EEE\}, \{SEE\}, \{ESE\}, \{SSE\}, \{EES, ESS, SES, SSS\}\}$$

$$T_U: \begin{matrix} \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ a'' & b'' & c'' & d'' & e'' \end{matrix} \quad \longleftrightarrow \Sigma_U$$

(b) The graph G_C for the three players is:



Thus $\Sigma_C = \{\emptyset, \Omega, C^1, C^2\}$ and the atoms are:

$$C^1 = \{EEE\}$$

$$C^2 = \Omega \setminus \{EEE\} = \{EES, ESE, ESS, SEE, SSE, SES, SSS\}$$

(c) The event "there is at least one economist" is $A = \Omega \setminus \{SSS\}$

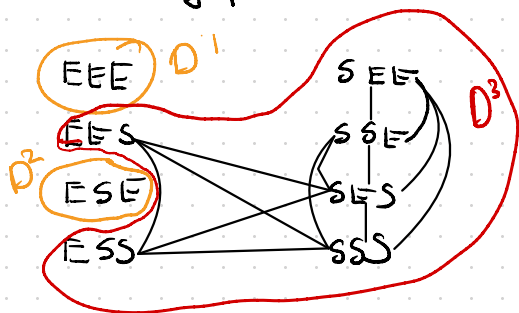
Given (b) this event is common knowledge only at $\{EEE\} = K_c A$

(d) The event "there is at least one sociologist" is $B = \Omega \setminus \{EEE\} = C^2$

So this event is common knowledge at all states $C^2 = K_c C^2$

(e) Suppose the state is ESE.

The graph G_c for Ahtohela & Uldana is:



So $\{ESE\} = D^2 \in \Sigma_A \cap \Sigma_U$ so it is common knowledge between Uldana & Ahtohela that they are economists and Bayan is a sociologist.

However, when adding Bayan, by the previous parts, $\{ESE\}$ is not common knowledge.

Problem 2

This problem is similar to the one's in the Beliefs list. I will use the same technique I used in the solutions to PSI to solve all items simultaneously:

Define $Q_1^0 = E[X|\Sigma_1]$, $Q_2^0 = E[X|\Sigma_2]$ and $Q_i^t = E[X|\Sigma_i, Q_j^0, Q_j^1, \dots, Q_j^{t-1}]$
for $i, j = 1, 2$, $t \geq 1$

We are going to compute the sequence of RV's Q_1^t, Q_2^t and then look at their realizations on $\omega = 5, 1, 4, 8, 9$ (that is, like if they were simultaneously announcing, like in the problem list).

$$Q_1^0 = \begin{cases} 1/3 & \omega \in \{1, 2, 3\} \\ 1/3 & \omega \in \{4, 5, 6\} \\ -1/3 & \omega \in \{7, 8, 9\} \end{cases}$$

$$Q_2^0 = \begin{cases} 0 & \omega \in \{1, 2, 3, 4\} \\ 0 & \omega \in \{5, 6, 7, 8\} \\ 1 & \omega = 9 \end{cases}$$

$$Q_1^1 = Q_2^1; \quad Q_2^1 = \begin{cases} 1/3 & \omega \in \{1, 2, 3\} \\ 1 & \omega = 4 \\ 1 & \omega \in \{5, 6\} \\ -1 & \omega \in \{7, 8\} \\ 1 & \omega = 9 \end{cases}$$

$$Q_1^1 = \begin{cases} 1/3 & \omega \in \{1, 2, 3\} \\ 1/3 & \omega \in \{4, 5, 6\} \\ -1 & \omega \in \{7, 8\} \\ 1 & \omega = 9 \end{cases}$$

$$Q_2^1 = \begin{cases} 0 & \omega \in \{1, 2, 3, 4\} \\ 1 & \omega \in \{5, 6\} \\ -1 & \omega \in \{7, 8\} \\ 1 & \omega = 9 \end{cases}$$

From the previous, we can determine what will be the sequence of "Yes", "No" responses:

• $\omega = 5$: P1 Yes $\rightarrow$ P2 No $\rightarrow$ End

• $\omega = 1$: P1 Yes $\rightarrow$ P2 NO $\rightarrow$ End

• $\omega = 4$: P1 Yes $\rightarrow$ P2 NO $\rightarrow$ End

• $\omega = 8$: P1 NO $\rightarrow$ END

• $\omega = 9$: P1 NO $\rightarrow$ END.

$$Q_1^2 = \begin{cases} 1/3 & \omega \in \{1, 2, 3\} \\ -1 & \omega = 4 \\ 1 & \omega \in \{5, 6\} \\ -1 & \omega \in \{7, 8\} \\ 1 & \omega = 9 \end{cases}$$

$$Q_2^2 = \begin{cases} 0 & \omega \in \{1, 2, 3, 4\} \\ 1 & \omega \in \{5, 6\} \\ -1 & \omega \in \{7, 8\} \\ 1 & \omega = 9 \end{cases}$$

Problem 3

(a) The belief space $N = \{1(\text{buyer}), 2(\text{seller})\}$ $\Omega = \{G, B\}$, $T_1 = \{a\}$, $T_2 = \Omega$

$$T_1(G) = T_1(B) = a, \quad T_2(G) = G, \quad T_2(B) = B, \quad \mu(G) = q, \quad \mu(B) = 1-q$$

(b) For Player 1: $S_1 = \{B(\text{buy}), D(\text{don't buy})\}$

For player 2; Let " s " denote the action sell, and D denote don't sell. Thus

$$S_2 = \{s_2: T_2 \rightarrow A_2\} = \{s_2: \{G, B\} \rightarrow \{s, D\}\}$$

if we represent the action taken at each of her types by the vector $(s_2(G), s_2(B))$ then

$$= \{(s, s), (s, D), (D, s), (D, D)\}$$

(c) We will consider only values for $p \geq 0$, since p is a price. Since there is complete information, we can analyze each matrix game separately.

We start analyzing the Seller:

When $q=1$, The seller

$\begin{cases} s \text{ is a weakly dominant strategy if } p > 5 \\ D \text{ is a weakly dominant strategy if } p < 5 \\ \text{when } p = 5, \text{ any } \{s, D\} \text{ are best responses against all of buyer's actions.} \end{cases}$

When $q=0$, the seller

$\begin{cases} s \text{ is weakly dominant if } p > 0 \\ \text{when } p = 0, \text{ any } \{s, D\} \text{ are best responses against all of buyer's actions.} \end{cases}$

Similarly, for the Buyer:

When $q=1$ $\left\{ \begin{array}{l} \text{Buyer has B as weakly dominant strategy if } p < 6 \\ \text{Buyer has D as weakly dominant strategy if } p > 6 \\ \{B, D\} \text{ is a best response to any action of the seller if } p = 6 \end{array} \right.$

When $q=0$ $\left\{ \begin{array}{l} \text{Buyer has B as weakly dominant strategy if } p < 4 \\ \text{Buyer has D as weakly dominant strategy if } p > 4 \\ \{B, D\} \text{ is a best response to any action of the seller if } p = 4 \end{array} \right.$

Combining all of the above:

When $q=1$ the equilibrium is: $\left\{ \begin{array}{l} \text{if } p < 5 \quad (B, D) \\ \text{if } p = 5 \quad (B, \downarrow) \& (B, D) \\ \text{if } 5 < p < 6 \quad (B, \downarrow) \\ \text{if } p = 6 \quad (B, \downarrow) \& (D, \downarrow) \\ \text{if } p > 6 \quad (D, \downarrow) \end{array} \right.$

When $q=0$ the equilibrium is: $\left\{ \begin{array}{l} \text{if } p = 0 \quad (B, \downarrow) \& (B, D) \\ \text{if } 0 < p < 4 \quad (B, \downarrow) \\ \text{if } p = 4 \quad (B, \downarrow) \& (D, \downarrow) \\ \text{if } p > 4 \quad (D, \downarrow) \end{array} \right.$

(d) Assume $q \in (0, 1)$. By the analysis from part (c), both seller types G & B want to sell if $p \geq 5$, i.e., $S_2^+(G) = S_2^+(B) = \downarrow$ if $p \geq 5$.

If this is the case for player 1 (Buyer) the expected payoffs are:

Buy: $q(6-p) + (1-q)(4-p)$ } Thus, the buyer never wants to buy if:
Don't: 0 } $q(6-p) + (1-q)(4-p) < 0 \Leftrightarrow q < \frac{p-4}{2}$

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 $\Rightarrow$

Thus if $p \geq 5$ AND $0 < q < \frac{p-4}{2}$ then the equilibrium is $(D, (4, 4))$

(e) Combining (c) & (d), the only equilibria left to find are:

• $p \geq 5, q > \frac{p-4}{2}$ then the equilibrium is $(B, (4, 4))$

• $p \geq 5, q = \frac{p-4}{2}$ then the equilibria is $(B, (4, 4)), (D, (4, 4))$

Now, note that if $p < 5$, Seller has a weakly dominant strategy if not selling when G , and as long as $p \geq 0$, Seller will sell when Bad, i.e., $S_2^*(G) = D, S_2^*(B) = D$. But then the expected payoff of the Buyer when buying & don't are, respectively:

B: $(1-q)(4-p)$ } $\Rightarrow$ If $p < 4$, buyer Buy

D: 0 } If $p > 4$ buyer doesn't buy

If $p = 4$, both are best responses.

Combining the above,

For any $q \in (0, 1)$: If $\begin{cases} 4 < p < 5 \\ p = 4 \\ 0 < p < 4 \\ p = 0 \end{cases}$ $\begin{cases} (D, (0, 4)) \text{ is an eq.} \\ (D, (0, 4)) \text{ \& } (B, (0, 4)) \text{ are equilibria} \\ (B, (0, 4)) \text{ is an equilibrium.} \\ (B, (0, 4)), (B, (0, 0)) \text{ are equilibria} \end{cases}$